NTRU with Hints: Recovering NTRU Secret Keys from Partial Leakage

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Abstract. NTRU-based structured lattices underpin several standardized post-quantum cryptographic schemes, most notably the Falcon signature algorithms. While offering compactness and efficiency, the algebraic structure of NTRU lattices introduces new vulnerabilities under physical attacks, where partial secret key leakage may occur.

This work addresses the problem of full key recovery in NTRU-based schemes when adversaries obtain partial information through side-channel or fault attacks. Existing leakage-aware frameworks, including the DDGR estimator and the approach of May and Nowakowski, either lack scalability or are limited to structured, single-source leakage on one secret vector. These constraints make them ineffective against practical leakage patterns in NTRU settings.

We propose a unified and scalable framework for recovering NTRU secret keys under partial leakage. Our method supports diverse hint types, such as perfect hints, modular hints, and low-bit leakage, and enables joint integration of leakage across both secret polynomials f and g. At its core, the framework uses a dimension-reduction strategy to eliminate known coefficients and reduce the problem to a lower-dimensional NTRU instance suitable for lattice reduction. Additionally, we introduce a transformation that converts hints on g into modular constraints on f, allowing unified hint embedding.

We demonstrate practical attacks on Falcon using NIST reference implementations. Leaking 400 coefficients of f in Falcon-512 reduces the required BKZ block size from over 350 to 38, enabling full key recovery within 6 hours. Compared to MN23, our method achieves significant speedups: 5.83× for Falcon-512 with 400 leaked coefficients, and over 15× for Falcon-1024 with 910 leaked coefficients. These results highlight the efficiency and scalability of our framework and the importance of leakage-resilient design for structured NTRU lattices.

Keywords: Post-quantum cryptography, NTRU lattice, Cryptanalysis, NTRU with hints, Lattice reduction

1 Introduction

The rapid development of quantum computing poses a significant threat to modern cryptographic infrastructures. Shor's algorithm [34], which efficiently solves integer factorization and discrete logarithm problems, compromises the security of widely used cryptosystems such as RSA and elliptic-curve cryptography (ECC). This motivates the need for cryptographic schemes that remain secure even in the presence of quantum adversaries.

Post-Quantum Cryptography (PQC) aims to construct quantum-resistant schemes, and lattice-based cryptography has emerged as a leading direction due to its theoretical hardness and implementation efficiency. In the NIST PQC standardization effort, three lattice-based schemes—Kyber [7], Dilithium [9], and Falcon [13]—have been standardized. Among them, Falcon is based on the NTRU lattice problem [20] and employs structured lattices over the negacyclic ring $\mathbb{Z}[X]/(X^n+1)$, enabling compact key and signature sizes. Falcon has been adopted in the Algorand blockchain [3] and Google Cloud's ALTS protocol [15] in hybrid key exchange mechanisms using NTRU-HRSS [23].

Threat Model. In realistic deployments, secret keys in NTRU-based schemes are composed of two small-coefficient polynomials f and g, typically sampled from discrete Gaussian or binomial distributions. Side-channel [24,16,35], fault injection [25,31], cold boot [2], and decryption failure [22] attacks can leak partial information about these polynomials—either selected coefficients or low-order bits. We assume an attacker has access to public keys and a moderate number of valid signatures, and can extract partial hints via physical attacks. The attacker then attempts to reconstruct the full secret key by solving a structured lattice problem informed by the leaked data.

Related Work. Previous research on lattice-based signatures has focused on both the design of secure instantiations and the cryptanalysis of their structural weaknesses. From a design perspective, the original NTRU cryptosystem [20] inspired many variants including NTRUSign [19], which was later broken due to its algebraic leakage [30]. The GPV framework [14] enabled provable security in hash-and-sign schemes, which was realized concretely in Falcon [13]. Falcon's successors, such as Mitaka [11] and Antrag [12], improved sampling efficiency and side-channel resistance.HAWK [10] further simplified the implementation by employing centered binomial sampling and introducing a new hardness assumption, namely the search module lattice isomorphism problem (smLIP).

In parallel, attacks against NTRU lattices have explored hybrid [21], subfield [1], and dense sublattice [26] techniques. Gama et al.'s conjecture that NTRU instances can reduce to SVP in halved dimension was confirmed in [6]. However, most of these efforts assume idealized security models and do not consider partial physical leakage.

To address this, Dachman-Soled et al. [8] proposed the leaky-LWE-Estimator framework, classifying side information as perfect, modular, approximate, and short vector hints. Each hint imposes a constraint on the target lattice, embedded into a Distorted Bounded Distance Decoding (DBDD) instance. While foundational, the sequential embedding of hints limits scalability. May and Nowakowski [29] improved efficiency by enabling batched hint integration, demonstrating full key recovery for Falcon-512 under carefully crafted perfect hints. However, their framework supports only secret-only leakage and assumes structured leakage on a single vector. Kirshanova et al. [27] introduced almost-parallel hints to generalize leakage scenarios, but their approach remains theoretical and computationally intensive in practice.

NTRU with Hints. In post-quantum cryptographic schemes based on NTRU lattices, the secret polynomials f and g may satisfy certain structural constraints, or partial information about them may be leaked through attack methods such as side-channel analysis. These constraints or leaked information are collectively referred to as hints.

A public LWE instance $(\mathbf{A}, \mathbf{b}, q) \in \mathbb{Z}_q^{n\times m} \times \mathbb{Z}_q^m \times \mathbb{N}$ satisfies the following LWE relation: $\mathbf{sA} + \mathbf{e} \equiv \mathbf{b} \mod q$, where $\mathbf{s} \in \mathbb{Z}_q^n$ is a secret vector with small norm, and $\mathbf{e} \in \mathbb{Z}_q^m$ is an error vector. The NTRU problem can be seen as a special case of the Ring-LWE problem where $\mathbf{b} = \mathbf{0}^m$.

According to [8,27], for the NTRU problem, an attacker may obtain several types of hint information related to the secret $\mathbf{f}$ and $\mathbf{g}$ or the lattice Λ. Let us assume the attacker knows the parameters $\mathbf{v}, \ell, k, \sigma, q$,

- Perfect hints: $\langle \mathbf{f}, \mathbf{v}\rangle = \ell$ define a hyperplane intersecting the lattice.
- Modular hints:$\langle \mathbf{f}, \mathbf{v}\rangle \equiv \ell \mod k$ sparsify the lattice.
- Approximate hints: $\langle \mathbf{f}, \mathbf{v}\rangle = \ell + \epsilon_\sigma$ reduce the covariance of the secret.
- Short vector hints: $\mathbf{v} \in \Lambda$ can be projected away from the lattice.
- Almost-parallel hint: $\mathbf{f} = \ell\mathbf{v} + \mathbf{f}'$. If the secret vector $\mathbf{f}$ is approximately parallel to a known vector $\mathbf{v}$, where $\mathbf{v}$ is not necessarily contained in the target lattice, then $\mathbf{f}$ can be decomposed in this form. Here, $\mathbf{f}'$ is a vector significantly shorter than $\mathbf{f}$.

A perfect leak of a secret coefficient $\mathbf{f}_i = \ell$ may occur in practice. Modular hints generalize this concept to modular relations, represented as linear equations modulo an integer or modulo q. These hints may originate from side-channel attacks on modular operations or integer ring computations. For example, during the preimage computation of Gaussian sampling in the Hawk signature scheme, side-channel leakage can reveal low-order bits of the secret $\mathbf{f}$, resulting in hints of the form $\mathbf{f}_i = \ell \bmod 2$. Similarly, hint leakage may also arise at the final stage of the NTT transformation. Approximate hints can be interpreted as noisy linear equations. For example, in the side-channel attack on Falcon proposed in [16], the so-called Hidden Parallelepiped structure emerges during the signing phase. As a result, the vector $(\mathbf{f}',\ \mathbf{g}')$ recovered through side-channel leakage and the HPP solver is an approximation of the true secret $(\mathbf{f},\ \mathbf{g})$. Short vector hints, while generally ineffective for direct key recovery, are designed to extend a standard technique that involves *ignoring* certain LWE equations. This form of relaxation can be interpreted geometrically as an orthogonal projection against a so-called q-vector. Projecting the lattice onto $\mathbf{v}$ significantly shortens the secret vector $\mathbf{f}$:

$\|\pi_{\mathbf{v}}(\mathbf{f})\| = \|\pi_{\mathbf{v}}(\mathbf{f}')\| \leq \|\mathbf{f}'\|$. Moreover, by integrating an almost-parallel hint, the dimension of the lattice can be reduced by one.

Despite these advances, no existing method supports joint integration of partial, heterogeneous hints across f and g, nor enables efficient full key recovery under realistic leakage for full-size NTRU schemes.

1.1 Motivation

While NTRU-based schemes such as Falcon have been standardized and adopted, their security under partial secret key leakage remains insufficiently explored. Existing analyses focus on idealized secrets over $\mathbb{Z}[X]/(X^n - 1)$, such as the ternary polynomials used in NTRU-HPS and NTRU-HRSS. However, practical schemes like Falcon and Mitaka operate over the irreducible negacyclic ring $\mathbb{Z}[X]/(X^n + 1)$, which induces different symmetry and short vector behavior. For example, NTRU-HPS exhibits cyclic symmetry with short vectors $(X^i f, X^i g)$, while Falcon has negacyclic vectors $((-1)^i X^i f, (-1)^i X^i g)$, which alters lattice hardness and leakage propagation.

In practice, attackers often obtain only a small fraction of the secret key coefficients or low-order bits, e.g., 100 coefficients or $f_i \bmod 2$. Existing methods like DDGR become inefficient beyond small n, and MN assumes uniform, full hint structures. Even in MN's reported success on Falcon-512 with 233 hints, the leakage is assumed to be carefully structured. Realistic side-channel traces typically yield fragmented, noisy hints that violate such assumptions.

Moreover, most frameworks treat f and g independently, neglecting the fact that both appear in signature generation and can leak simultaneously. A unified recovery strategy must integrate cross-component hints. Additionally, existing tools focus on security estimation, but practical key recovery in concrete parameter sets remains an open challenge.

These gaps highlight the need for a unified, scalable attack framework that supports partial, asymmetric, and joint leakage across both f and g, and enables practical key recovery in structured lattice schemes under real-world conditions.

1.2 Our Contributions

We propose a unified, efficient attack framework for structured NTRU-based schemes under partial secret key leakage. Our framework accommodates realistic leakage patterns and scales to full-size parameter sets.

Key Recovery Method via Dimension Reduction. We introduce a dimension-reduction strategy that eliminates known coefficients before lattice construction, avoiding sublattice embedding. Our attack on Falcon-512 with 400 leaked coefficients reduces the BKZ block size from over 350 to approximately 38, enabling full key recovery within 6 hours on a modern workstation.

Unified Hint Integration across f and g. We develop a transformation that converts perfect hints on g into modular constraints on f, enabling cross-component hints to be embedded in a unified lattice basis. This supports mixed hint types from both secrets, improving information utilization.

Theoretical and Empirical Analysis of Hint Types. We analyze how different hint types—including perfect, modular, and mod-2—affect lattice volume and recovery hardness. Based on core-SVP models [4], we quantify security degradation and determine thresholds where key recovery becomes feasible.

Experimental Validation. We implement full key recovery attacks on Falcon and Mitaka using NIST reference code. Compared to MN [29], Our method achieves significant improvements in clocktime compared to MN23. For Falcon-512 with 400 leaked coefficients, our method is approximately 5.83× faster, and for Falcon-1024 with 910 leaked coefficients, it is over 15× faster. These results highlight the superior efficiency and scalability of our approach in high-dimensional NTRU settings.

2 Preliminaries

2.1 Notation

Let $\phi = X^n + 1$ for $n = 2^\kappa$, a power of two, and $q \in \mathbb{N}^*$. We denote by $\mathbb{R} = \mathbb{Z}[x]/(X^n + 1)$ and $\mathbb{R}_q = \mathbb{Z}_q[x]/(X^n + 1)$.

Scalars are written in regular font. Bold lower-case letters denote vectors with coefficients in R or R_q and are assumed to be row vectors by default. Bold upper-case letters denote matrices. For a vector $\mathbf{b}$, we denote its transpose by $\mathbf{b}^{\mathrm{T}}$. For a matrix $\mathbf{A}$ with row vectors $\mathbf{a}_1, \ldots, \mathbf{a}_n$, we write $\mathbf{A} = \left[\mathbf{a}_1^{\mathrm{T}}; \cdots; \mathbf{a}_n^{\mathrm{T}}\right]$ Its transpose is denoted by $\mathbf{A}^{\mathrm{T}}$.

The i-th standard basis vector of $\mathbb{R}^n$ is denoted by $\mathbf{e}_i$ (e.g., $\mathbf{e}_1 = (1, 0, \ldots, 0)$). The n-dimensional identity matrix is denoted by $\mathbf{I}_n$, the all-zero $(n \times m)$ matrix by $\mathbf{0}_{n\times m}$, and the n-dimensional all-zero vector by $\mathbf{0}^n$. When dimensions are clear from context, we omit the subscripts.

A polynomial $f(x) = f_0 + f_1 x + \cdots + f_{n-1}x^{n-1}$ in R_q is identified with its coefficient vector $\mathbf{f} = (f_0, f_1, \ldots, f_{n-1})$, treated as a row vector. The Euclidean norm and inner product are denoted by $\|\cdot\|$ and $\langle\cdot,\cdot\rangle$, respectively.

2.2 Lattices

Definition 2.1 (Lattice). A *lattice*, denoted as Λ, is a discrete additive subgroup of $\mathbb{R}^m$, which is generated as the set of all linear integer combinations of n $(m \geq n)$ linearly independent basis vectors $\{\mathbf{b}_j\} \subset \mathbb{R}^m$, namely,

$$\Lambda := \left\{ \sum_j z_j \mathbf{b}_j \ : \ z_j \in \mathbb{Z} \right\} \tag{1}$$

We say that m is the *dimension* of Λ and n is its *rank*. A lattice is *full rank* if $n = m$. A matrix $\mathbf{B}$ having the basis vectors as rows is called a *basis*. The *volume* of a lattice Λ is defined as

$$\mathrm{Vol}(\Lambda) := \sqrt{\det(\mathbf{B}\mathbf{B}^T)}. \tag{2}$$

The *dual lattice* of Λ in $\mathbb{R}^n$ is defined as follows:

$$\Lambda^* := \{\mathbf{y} \in \mathrm{Span}(\mathbf{B}) \mid \forall \mathbf{x} \in \Lambda,\ \langle \mathbf{x}, \mathbf{y} \rangle \in \mathbb{Z}\}.$$

Note that, $(\Lambda^*)^* = \Lambda$, and $\mathrm{Vol}(\Lambda^*) = 1/\mathrm{Vol}(\Lambda)$.

2.3 LWE Problem

Definition 2.2 (Search LWE problem with short secrets). Let n, m and q be positive integers, and let χ be a distribution over $\mathbb{Z}$. The LWE problem with short secrets for parameters (n, m, q, χ) is defined as follows: Given

- a uniformly random matrix $\mathbf{A} \in \mathbb{Z}_q^{n \times m}$, and
- a vector $\mathbf{b} \equiv \mathbf{s}\mathbf{A} + \mathbf{e} \mod q$, where $\mathbf{s} \leftarrow \chi^n, \mathbf{e} \leftarrow \chi^m$,

Find $\mathbf{s} \in \mathbb{Z}_q^n$. The vector $\mathbf{s}$ is the secret, $\mathbf{e}$ is the error. The tuple $(\mathbf{A}, \mathbf{b}, q)$ is called an LWE instance.

While Regev's original formulation of LWE [32] assumes that the secret and the error follow distinct distributions, the majority of real-world LWE-based cryptographic constructions adopt the short secret variant, as formalized in Definition 2.2, due to its implementation efficiency and security.

2.4 NTRU problem

Definition 2.3 (NTRU [20]). Let n, q be positive integers, and let $f, g \in \mathbb{Z}_q[x]$ be polynomials of degree n sampled from some distribution χ, subject to f being invertible modulo a polynomial φ of degree n. Define

$$h = \frac{g}{f} \mod q. \tag{3}$$

The *NTRU problem* is the problem of finding f, g given h (or any equivalent solution $(x^i \cdot f,\ x^i \cdot g)$ for some $i \in \mathbb{Z}$).

The NTRU problem can be interpreted as a special case of the LWE problem. Specifically, the standard LWE problem is given by: $\mathbf{b} \equiv \mathbf{s}\mathbf{A} + \mathbf{e} \mod q$,

The NTRU problem can be viewed as a particular instance of LWE, where the negacyclic matrix generated from the public key polynomial h is denoted by $\mathbf{H}_h$, and the polynomial representations of f and g are expressed in vector form. Then, the corresponding LWE system becomes:

$$\mathbf{b} = \mathbf{0}^m, \quad \mathbf{A} = \mathbf{H}_h, \quad \mathbf{s} = \mathbf{f}, \quad \mathbf{e} = -\mathbf{g}.$$

Therefore, solving the NTRU problem is equivalent to solving an LWE instance with $\mathbf{b} = \mathbf{0}^m$. This demonstrates that NTRU can be regarded as a structured variant of the Ring-LWE problem, where recovering the secrets $\mathbf{f}$ and $\mathbf{g}$ amounts to solving the following equation:

$$\mathbf{f} \cdot \mathbf{H}_h - \mathbf{g} \equiv \mathbf{0} \mod q. \tag{4}$$

2.5 Lattice Reduction

Any lattice of dimension at least two admits infinitely many different bases. In practical applications such as cryptanalysis, one aims to find a good basis, i.e., one composed of short and nearly orthogonal vectors. The LLL algorithm [28] computes a relatively good basis in polynomial time, although the length of its shortest vector remains exponentially larger than that of the true shortest vector in the lattice. The BKZ algorithm [33,5], as a generalization of LLL, achieves a better trade-off between clocktime and basis quality. The most critical parameter in BKZ is the blocksize β, which determines the dimension of the projected sublattices on which the shortest vector is computed. Since most of the BKZ clocktime is spent on this step, β dominates the total clocktime. In practice, the security of lattice-based cryptographic schemes is usually measured by the smallest blocksize β needed to solve the target instance. Estimating the required blocksize β for a given lattice is an important and active research problem.

2.6 BDD Problem

Bounded Distance Decoding (BDD) is a special case of the Closest Vector Problem (CVP).

Given a lattice $\mathcal{L} \subset \mathbb{R}^n$ and a target vector $\mathbf{v} \in \mathbb{R}^n$, the target vector satisfies

$$\mathrm{dist}(\mathbf{v}, \mathcal{L}) < \gamma \lambda_1(\mathcal{L}),$$

where the goal is to find the closest lattice point $\mathbf{u} \in \mathcal{L}$ to the target vector $\mathbf{v}$.

Compared to the CVP problem, BDD imposes an additional constraint. BDD is a variant of CVP, but the difference is that BDD assumes that the target vector $\mathbf{v}$ is already very close to some lattice point $\mathbf{u}$, and the solution must ensure that the distance between the target vector and the closest lattice point is within a known bounded range.

2.7 Heuristic

Heuristic 2.1 (Gaussian Heuristic). [8] Let Λ be an n-dimensional lattice. The Gaussian heuristic predicts that $\lambda_1(\Lambda)$ equals

$$\mathrm{gh}(\Lambda) := \sqrt{\frac{n}{2\pi e}} \det(\Lambda)^{1/n} \tag{5}$$

A set of linearly independent lattice vectors $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\} \subset \Lambda$ is called primitive (with respect to Λ), if it can be extended to a basis of Λ. Equivalently, the set $\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}$ is called primitive, if

$$\Lambda \cap \operatorname{span}_{\mathbb{R}}(\{\mathbf{v}_1, \ldots, \mathbf{v}_k\}) = \mathcal{L}(\mathbf{v}_1, \ldots, \mathbf{v}_k).$$

Heuristic 2.2. [29] *Let $q \in \mathbb{N}$ be an odd integer and let*

$$M_{k,q} := \left\{ \mathbf{B} \in \mathbb{Z}^{k\times k} \;\middle|\; \det \mathbf{B} \neq 0,\, \mathbf{B} \text{ has entries in } \left\{ -\frac{q-1}{2}, \ldots, \frac{q-1}{2} \right\} \right\}.$$

Let $\mathbf{B}$ be uniformly distributed over $M_{k,q}$. Then it holds that

$$\mathbb{E}[\ln(|\det \mathbf{B}|)] \sim \frac{k}{2} \ln \left(\frac{q^2-1}{12e} \cdot k \right) \tag{6}$$

as $k \to \infty$.

2.8 The Primal Lattice Reduction Attack[29]

For an instance of the LWE $(\mathbf{A}, \mathbf{b}) \in \mathbb{Z}_q^{n\times m} \times \mathbb{Z}_q^m$ that satisfies

$$\mathbf{sA} + \mathbf{e} \equiv \mathbf{b} \mod q,$$

where $\mathbf{s} \in \mathbb{Z}_q^n$ and $\mathbf{e} \in \mathbb{Z}_q^m$. The q-ary lattice corresponding to the row vectors of matrix $\mathbf{A}$ is given by

$$q\text{-ary}(\mathbf{A}) = \begin{pmatrix} q\mathbf{I}_m & 0 \\ \mathbf{A} & \mathbf{I}_n \end{pmatrix}.$$

By viewing the LWE problem as a Bounded Distance Decoding (BDD) problem on the q-ary lattice q-ary$(\mathbf{A})$ and embedding the vector $\mathbf{b}$, we obtain the LWE lattice Λ^{LWE}:

$$\mathbf{B}^{\text{LWE}} = \begin{pmatrix} q\mathbf{I}_m & 0 & 0 \\ \mathbf{A} & \mathbf{I}_n & 0 \\ \mathbf{b} & 0 & 1 \end{pmatrix},$$

where the row vectors form the lattice basis.

$$\Lambda^{\text{LWE}} := \{(\mathbf{x}, \mathbf{y}, t) \in \mathbb{Z}^n \times \mathbb{Z}^m \times \mathbb{Z} \mid \mathbf{x} \equiv \mathbf{yA} + t\mathbf{b} \mod q\}.$$

The target vector

$$\mathbf{t} := (-\mathbf{e}, \mathbf{s}, -1) \in \Lambda^{\text{LWE}}.$$

3 Integrating PSCL of secrect as Perfect Hints in NTRU Instance

This section focuses on the security impact of Partial Secret Coefficient Leakage (PSCL) of the secret polynomial f in NTRU instances, and investigates how to recover the unknown coefficients under such leakage. In most existing NTRU-based cryptographic schemes, the secret polynomials f and g are typically sampled from the same distribution. Therefore, we consider both the case of partial coefficient leakage from f alone and the case where partial coefficients of both f and g are leaked. We interpret such leakage as a form of perfect hint. Specifically, we analyze in detail how different choices of the vector $\mathbf{v} \in \mathbb{Z}^n$ in the perfect hint relation $\langle \mathbf{f}, \mathbf{v} \rangle = \ell$ affect the volume of the constructed lattice, evaluate the security degradation resulting from partial coefficient leakage, and propose a practical attack method with significantly reduced computational overhead.

3.1 Impact of Hint Vector Construction on Lattice Volume and Security in Practice

In the experimental setting of [29], since LWE systems often involve inner products between the secret and the hint vectors, the hint vectors are typically assumed to be approximately random. However, this assumption no longer holds when the leakage corresponds to partial coefficients of the secret key.

Specifically, in [29], each hint vector $\mathbf{v}_i \in \mathbb{Z}_q^n$ is uniformly sampled from the set $\{0, \ldots, q-1\}^n$. Given $\ell_i = \langle \mathbf{f}, \mathbf{v}_i \rangle$, where $\mathbf{f}$ is the secret key vector, the attacker is assumed to possess k perfect hints of the form $\bar{\mathbf{v}}_i = (\mathbf{v}_i, \ell_i)$. The hint matrix is then constructed as: $\mathbf{H} := \mathrm{Hint}(\bar{\mathbf{v}}_1, \ldots, \bar{\mathbf{v}}_k)$.

Each perfect hint introduces an additional linear constraint, thereby increasing the volume of the NTRU-type lattice Λ^{NTRU}, with the growth in determinant proportional to $|\det \mathbf{H}|$. According to Heuristic 2.2 in [29], the following relation holds:

$$\det \Lambda^{\mathrm{NTRU}}_{\mathbf{H},k} = \det \Lambda^{\mathrm{NTRU}} \cdot \det \mathcal{L}(\bar{\mathbf{v}}_1, \ldots, \bar{\mathbf{v}}_k) = \det \Lambda^{\mathrm{NTRU}} \cdot \det \mathcal{L}(\mathbf{H}).$$

In NTRU-based cryptographic schemes, the hint vectors $\mathbf{v}_i$ used to model information leakage are typically categorized into the following two types:

- **Type I (Random Hint Vector in $\{0, \ldots, q-1\}^n$):** As shown in the experimental setup of [29], each $\mathbf{v}_i$ is chosen uniformly at random from $\{0, \ldots, q-1\}^n$. This models an unconstrained linear combination with the secret vector $\mathbf{f}$. In this setting, the squared norm of $\ell_i = \langle \mathbf{f}, \mathbf{v}_i \rangle$ is large, leading to significant growth in lattice volume. Given k perfect hints $\bar{\mathbf{v}}_i = (\mathbf{v}_i, \ell_i)$, solving for short vectors using the primal lattice reduction attack becomes relatively easier in this setting, because the determinant of Λ^{LWE} increases by a factor of roughly

$$\exp\left(\frac{k}{2} \ln\left(\frac{q^2}{12e} \cdot k\right)\right) = \left(\sqrt{\frac{k}{12e}} \cdot q\right)^k \tag{7}$$

- **Type II (Perfect Hints Constructed from Standard Basis Vectors $\{\mathbf{e}_j\}_{j=1}^n$):** This setting is more constrained and closely aligned with practical

attack scenarios, where each hint vector takes the form $\mathbf{v}_i = \mathbf{e}_{j_i}$, a standard basis vector with a 1 in the j_i-th position and zeros elsewhere. In this case, the leaked value becomes $\ell_i = \langle \mathbf{f}, \mathbf{v}_i \rangle$, directly revealing one coefficient of the secret key f.
Let $\mathbf{V} \in \mathbb{Z}^{k \times n}$ denote the matrix whose rows are standard basis vectors $\mathbf{v}_i = \mathbf{e}_{j_i}$, and let $\boldsymbol{\ell} = (\ell_1, \ldots, \ell_k)^\top$ be the corresponding vector of inner products. The perfect hint matrix is then constructed as:

$$\mathbf{H} = [\mathbf{V} \mid \boldsymbol{\ell}] \in \mathbb{Z}^{k \times (n+1)}.$$

When the indices $\{j_1, \ldots, j_k\}$ are all distinct, the matrix $\mathbf{V}$ becomes a submatrix of the identity matrix, and thus its row vectors are linearly independent. As a result, the lattice generated by $\mathbf{V}$ has determinant:

$$\det(\mathcal{L}(\mathbf{v}_1, \ldots, \mathbf{v}_k)) = 1.$$

Furthermore, since $\boldsymbol{\ell}$ is completely determined by $\mathbf{V}$ and $\mathbf{f}$, the last column of $\mathbf{H}$ introduces no additional degrees of linear freedom. Consequently, the lattice volume remains unchanged:

$$\det\left(\mathcal{L}(\bar{\mathbf{v}}_1, \ldots, \bar{\mathbf{v}}_k)\right) = \det\left(\mathcal{L}(\mathbf{v}_1, \ldots, \mathbf{v}_k)\right) = 1. \tag{8}$$

This represents the most structured form of perfect hint leakage, where each hint imposes a strict algebraic constraint on the secret key without enlarging the lattice volume. Therefore, incorporating such perfect hints in the Primal Lattice Reduction Attack leads solely to a reduction in dimension, without increasing the determinant.

Different constructions of the hint vector v_i have a fundamental impact on the determinant of the resulting lattice, thereby influencing the success probability and computational cost of key recovery attacks. In the Type I setting, integrating perfect hints increases the lattice volume, which facilitates the primal lattice reduction attack by making short vector recovery easier. In contrast, in the Type II setting, the integration of perfect hints does not alter the lattice volume.

3.2 Evaluating the Impact of PSCL on Residual Security

A perfect hint of the form $\langle \mathbf{f}, \mathbf{v} \rangle = \ell$ provides strong auxiliary information in lattice-based cryptanalysis. For example, the NTRU-HPS scheme uses ternary secret vectors—where each element takes values from $\{-1, 0, 1\}$—and enforces a fixed number of ± 1 coefficients. This structure directly leads to the existence of one or two perfect hints. In the Falcon signature scheme, side-channel attacks have been shown to leak partial coefficients of the secret key f [24].

This raises an important question: how can we quantitatively evaluate the relationship between the number of leaked coefficients of f and the resulting residual security?

The work of [8] addresses this problem by proposing a method for integrating perfect hints into a DBDD instance. The core idea involves recursively intersecting the original LWE lattice with a sequence of orthogonal constraints, such as $\mathbf{e}_{m+i}^{\perp}$, to incrementally build the sublattice $\Lambda_{\mathbf{H},k}^{\mathrm{LWE}}$. At each step, the sublattice Λ_i is defined as the intersection of the previous sublattice Λ_{i-1} with the hyperplane orthogonal to $\mathbf{e}_{m+i}$, until all k constraints are incorporated. This methodology was proposed as a means of evaluating the security loss induced by side-channel leakage.

Although theoretically polynomial-time, this procedure is practically inefficient due to its inherently sequential nature—each step depends on the result of the previous one. This computational bottleneck is a major reason why the method in [8] is not well-suited for real-world cryptanalytic attacks. To overcome these limitations, follow-up work such as [29] proposed more efficient alternatives that bypass the costly sequential slicing process. These improvements enable more scalable and parallelizable attack strategies for high-dimensional LWE instances where perfect hints are available.

This experiment investigates, for the case of dimension $n = 512$, the minimum number of leaked coefficients of the secret key f or g in the Falcon signature scheme—referred to as perfect hints—required to enable a practically feasible attack. The evaluation employs the *leaky-LWE-Estimator* introduced in [8] to quantitatively analyze the effect of embedding perfect hints on the residual lattice dimension and the corresponding reduction in the required BKZ block size. This experiment investigates, for the case of dimension $n = 512$, the minimum number of leaked coefficients of the secret key f or g in the Falcon signature scheme—referred to as perfect hints—required to enable a practically feasible attack. The evaluation employs the *leaky-LWE-Estimator* introduced in [8] to quantitatively analyze the effect of embedding perfect hints on the residual lattice dimension and the corresponding reduction in the required BKZ block size.

For Falcon-512 and Falcon-1024, our evaluation approach is to regard the NTRU problem as an instance of the RLWE problem, and then use the leaky-LWE-Estimator from [8] for security evaluation. Since the method in [8] is based on the Primal Lattice Reduction Attack, the dimension in the case without any integrated hint information is $m + n + 1$. For Falcon-512, the dimension without any integrated hint information is 1025, with an estimated block size of $\beta = 350.46$. For Falcon-1024, the dimension without hint integration is 2049, with $\beta = 815.98$. When both the perfect hints from known partial coefficients and the inherent short vector hints of the lattice are integrated, the remaining security strength, reduced dimension d, and the corresponding changes are presented in Table 1 and Table 2.

According to the evaluation results, for Falcon-512, if 400 coefficients of the secret polynomial f are leaked, the residual security strength is reduced to $11.06\,\beta$, representing a degradation of $339.4\,\beta$. For Falcon-1024, if 905 coefficients of f are leaked, the residual security strength is $16.08\,\beta$, corresponding to a reduction of $799.9\,\beta$. The results for other leakage levels and the corresponding residual security strengths are presented in Table 1 and Table 2. It should be

Table 1: Impact of the coefficient leakage of f on Dimension and β in Falcon-512

Perfect Hints	400	405	410	415	420	425	430	435	440
Dimension d	260	246	232	217	200	189	178	167	157
BKZ Blocksize β	11.06	8.49	5.78	3.89	2.00	2.00	2.00	2.00	2.00

Table 2: Impact of the coefficient leakage of f on Dimension and β in Falcon-1024

Perfect Hints	905	910	915	920	925	930	935	940	945	950
Dimension d	278	265	252	238	222	206	193	182	172	161
BKZ Blocksize β	16.08	12.48	9.34	7.03	4.47	2.51	2.00	2.00	2.00	2.00

noted that these estimates may differ from the actual residual security observed in practical attacks. In the next section, we will focus on analyzing the residual security strength in real-world attack scenarios.

3.3 Integrating Partial Coefficients f into NTRU instance

In this section, we propose a novel algorithm 1 called Dim-and-Sample Reduced NTRU Attack (DSRNA), which aims to transform a high-dimensional NTRU instance with a large number of samples into a lower-dimensional instance with fewer samples by combining partial coefficient leakage of the secret key and sample elimination techniques. The reduced instance is then subjected to BKZ lattice basis reduction to recover the complete secret key.

An NTRU instance can be represented as $(\mathbf{H}_h, \mathbf{b} = \mathbf{0}^m, q)$, where $\mathbf{H}_h$ is the negacyclic matrix constructed from the public key h. The core idea of our method is to eliminate the NTRU samples corresponding to known secret key coefficients, thereby reducing the dimension of the primal lattice reduction attack and lowering the required BKZ block size. When calling the BKZ algorithm, it is feasible to reduce the number of samples used, as BKZ automatically determines the optimal number of samples. In fact, as pointed out in [8], when the number of samples becomes too large, the estimator may overestimate the required block size; however, in practice, the actual block size required does not increase accordingly. This indicates that increasing the number of samples does not improve the success rate of the attack. Therefore, our proposed method employs the leaky-LWE-Estimator to evaluate the optimal number of samples m' required for a successful primal attack, in order to ensure the minimal required BKZ block size and significantly reduce the time complexity of the subsequent attack.

The goal of this attack is to evaluate the impact on security caused by partial knowledge of the secret polynomial f, and to further recover the unknown coefficients of f. Once f is successfully recovered, the corresponding polynomial

g can be computed from the public key h, thereby enabling the reconstruction of the secret key sk.

Algorithm 1 Dim-and-Sample Reduced NTRU Attack

Require: An NTRU instance defined by $(\mathbf{H}_h, \boldsymbol{b}, q)$, where $\mathbf{H}_h \in \mathbb{Z}_q^{n\times m}$ and $\boldsymbol{b} \in \mathbb{Z}_q^m$; known part of the secret vector $\boldsymbol{f} = (\boldsymbol{f}^{[1:k]}, \boldsymbol{f}^{[k+1:n]})$; a sample size $m' \le m$.
Ensure: A candidate for the unknown secret part $\boldsymbol{f}' \in \mathbb{Z}^{n-k}$, or $\perp$ on failure.

1: **Step 1: Residual computation.**
2: Compute the residual target vector:

$$\boldsymbol{b}' := \left(\boldsymbol{b} - \boldsymbol{f}^{[1:k]} \cdot \left(\mathbf{H}_h^{[:,1:k]}\right)^{\mathrm{T}}\right) \bmod q.$$

3: **Step 2: Sample truncation.**
4: Let $\mathbf{H}'_h := \mathbf{H}_h^{[1:m',\, k+1:n]} \in \mathbb{Z}_q^{m'\times(n-k)}$ and $\boldsymbol{b}'' := \boldsymbol{b}'^{[1:m']} \in \mathbb{Z}_q^{m'}$.

5: **Step 3: Construct the extended lattice basis.**
6: Define a $(m'+n-k+1)$-dimensional lattice basis:

$$\mathbf{B}^{\mathrm{NTRU}} := \begin{pmatrix} q\cdot \mathbf{I}_{m'} & 0 & 0 \\ \mathbf{H}'_h & \mathbf{I}_{n-k} & 0 \\ \boldsymbol{b}'' & 0 & 1 \end{pmatrix}.$$

7: **Step 4: Lattice reduction.**
8: Apply BKZ basis reduction on $\mathbf{B}^{\mathrm{NTRU}}$ to find a short vector $\boldsymbol{t} = (\boldsymbol{g}, \boldsymbol{f}', -1) \in \mathbb{Z}^{m'+n-k+1}$.

9: **Step 5: Verification.**
10: Accept $\boldsymbol{f}'$ if it satisfies the original NTRU equation:

$$\boldsymbol{f}' \cdot \left(\mathbf{H}_h^{[1:m',\, k+1:n]}\right)^{\mathrm{T}} \equiv \boldsymbol{b}'^{[1:m']} \pmod q.$$

11: **if** the condition holds **then**
12:     **return** $\boldsymbol{f}'$
13: **else**
14:     **return** $\perp$
15: **end if**

The goal of this attack is to reconstruct the complete secret key sk by recovering the unknown coefficients of the secret polynomial f. To this end, we apply lattice basis reduction techniques to identify short vectors corresponding to the secret. Once f is successfully recovered, the corresponding polynomial g can be efficiently derived from the public key h, thus enabling full reconstruction of the secret key sk.

Table 3 presents the reduced dimension of the NTRU lattice after eliminating the known coefficients of the secret polynomial f, denoted by n'. The parameter m' represents the minimum number of samples required for a successful attack. The dimension of the primal attack lattice is computed as $m' + n' + 1$. We then use the *leaky-LWE-Estimator* tool to evaluate the residual security, particularly focusing on the required BKZ block size.

For example, when the lattice dimension is 72 and the number of samples is 84, the BKZ block size required to successfully perform the primal lattice reduction attack is 2.

To validate the effectiveness of the proposed DSRNA method, we experimentally evaluate the required BKZ block size and the corresponding time complexity for actual key recovery under secret coefficient leakage in Falcon-512, Falcon-1024, and Mitaka-512. The detailed results are provided in Section 5.

Table 3: Dim-and-Sample Reduced NTRU Attack for Falcon-512

Perfect hints	380	385	390	395	400	405	410	415	420
Dimension n_1	132	127	122	117	112	107	102	97	92
Number of samples m_1	175	169	163	150	147	138	129	119	107
Dimension of Primal Attack	308	297	286	268	260	246	232	217	200
BKZ Blocksize (β)	27.38	23.11	19.20	14.39	11.06	8.49	5.78	3.89	2.00

3.4 Integrating PSCL of f and g as perfect and Mod-q Hints in NTRU Instances

In practical attacks on NTRU-based cryptosystems, partial coefficient leakage may occur simultaneously for both $\mathbf{f}$ and $\mathbf{g}$, where $\mathbf{f}, \mathbf{g}$ are assumed to be *row vectors*. A natural question arises: how can we fully leverage the leakage information from both $\mathbf{f}$ and $\mathbf{g}$ to improve the success rate of secret key recovery?

The method proposed in [29] supports the integration of *secret-only* leakage—i.e., it can incorporate hints from either $\mathbf{f}$ or $\mathbf{g}$, but not both simultaneously. In this subsection, we extend the approach of [29] to enable the joint integration of partial coefficient leakage from both $\mathbf{f}$ and $\mathbf{g}$.

Suppose perfect hints on $\mathbf{f}$ and $\mathbf{g}$ are available, i.e., there exist vectors $\mathbf{v}$ and $\mathbf{w}$ such that

$$\langle \mathbf{f}, \mathbf{v} \rangle = l_1, \quad \langle \mathbf{g}, \mathbf{w} \rangle = l_2.$$

To jointly integrate leakage information from both $\mathbf{f}$ and $\mathbf{g}$, we propose converting the perfect hint on $\mathbf{g}$ into a *mod-q* hint on $\mathbf{f}$. Since in the NTRU setting we have

$$\mathbf{f} \cdot \mathbf{H}_h \equiv \mathbf{g} \pmod q$$

and

$$\langle \mathbf{g}, \mathbf{w} \rangle = l_2,$$

we can derive the following relation:

$$\langle \mathbf{f}, \mathbf{H}_h \cdot \mathbf{w} \rangle \equiv l_2 \pmod q.$$

In summary, if we possess a perfect hint on $\mathbf{f}$:

$$\langle \mathbf{f}, \mathbf{v} \rangle = l_1,$$

alongside a mod-q hint derived from the perfect hint on $\mathbf{g}$:

$$\langle \mathbf{f}, \mathbf{H}_h \cdot \mathbf{w} \rangle \equiv l_2 \pmod q,$$

then the latter can be viewed as a special case of a Modular Hint.

Consequently, the modular hint integration method proposed in [29] can be directly applied to embed both types of hint information into the NTRU lattice, thereby enabling a more effective Primal Lattice Reduction Attack.

Let $\mathbf{w}_1 = \mathbf{H}_h \cdot \mathbf{w}$. In this setting, for the perfect hint corresponding to $\mathbf{f}$, we define $\mathbf{v}_i = (\mathbf{v}_i, l_{1i})$ for $i = 1, \ldots, k$. Meanwhile, $\mathbf{w}_{1i} = (\mathbf{w}_i, l_{2i}, q)$ for $i = k+1, \ldots, k+\ell$.

By leveraging the method of modular and perfect hint integration proposed in [29], it is possible to combine both types of hints $\mathbf{v}_i$ and $\mathbf{w}_{1i}$ into a unified framework. However, [29] did not experimentally evaluate the attack performance of this approach. Therefore, we evaluate the attack effectiveness of this method specifically targeting Falcon signatures.

To this end, we construct two hint matrices, denoted as $H_1 := \mathsf{Hint}(\mathbf{v}_1, \ldots, \mathbf{v}_k) = (\mathbf{V}, \mathbf{l}_1)$ and $H_2 := \mathsf{Hint}(\mathbf{w}_{k+1}, \ldots, \mathbf{w}_{k+\ell}) = (\mathbf{W}, \mathbf{l}_2, \mathbf{Q})$, where

$$\mathbf{Q} := \begin{pmatrix} q & 0 & \cdots & 0 \\ 0 & q & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & q \end{pmatrix}$$

The process of embedding hints is equivalent to constructing an embedding matrix. Therefore, we can first embed the perfect hints followed by the modular hints, or alternatively embed the modular hints first and then the perfect hints — both orders are acceptable, since each hint is linearly independent.

Thus, embedding into the NTRU lattice Λ^{NTRU} results in a newly constructed lattice $\Lambda^{\mathsf{NTRU}}_{M,H}$, where $\mathbf{H}_h$ denotes the negacyclic matrix constructed from the public key. The corresponding lattice basis is:

$$\mathbf{B}^{\mathrm{NTRU}} := \begin{pmatrix} q\mathbf{I}_m & 0 & 0 \\ \mathbf{H}_h & \mathbf{I}_n & 0 \\ \mathbf{0}^m & 0 & 1 \end{pmatrix} \quad \rightarrow \quad \mathbf{B}^{\mathrm{NTRU}}_{M,H} := \begin{pmatrix} q\mathbf{I}_m & 0 & 0 & 0 & 0 \\ 0 & 0 & \mathbf{Q} & 0 & 0 \\ \mathbf{H}_h & \mathbf{V} & \mathbf{W} & \mathbf{I}_n & 0 \\ \mathbf{0}^m & \mathbf{l}_1 & \mathbf{l}_2 & 0 & 1 \end{pmatrix}$$

For the NTRU problem, there exists another type of perfect hint that involves simultaneous leakage of both $\mathbf{f}$ and $\mathbf{g}$, which is also commonly encountered in practical attack scenarios. Suppose we obtain a perfect hint of the form:

$$\big\langle (\mathbf{v}, \mathbf{w}), (\mathbf{f}, \mathbf{g}) \big\rangle = l \iff \mathbf{f} \cdot \mathbf{v}^{\mathrm{T}} + \mathbf{g} \cdot \mathbf{w}^{\mathrm{T}} = l. \tag{9}$$

then the question arises: how can this type of hint be embedded into the NTRU lattice used in the primal attack?

We begin by expressing $\mathbf{g}$ as a function of $\mathbf{f}$:

$$\mathbf{g} = (l - \mathbf{f}\mathbf{v}^{\mathrm{T}}) \cdot (\mathbf{w}^{\mathrm{T}})^{-1} = (l - \mathbf{f}\mathbf{v}^{\mathrm{T}}) \cdot (\mathbf{w}^{-1})^{\mathrm{T}}$$

Given that $\mathbf{f} \cdot \mathbf{H}_h \equiv \mathbf{g} \pmod{q}$, it follows that:

$$\mathbf{f} \cdot \mathbf{H}_h \equiv \left(l - \mathbf{f}\mathbf{v}^{\mathrm{T}}\right)\left(\mathbf{w}^{-1}\right)^{\mathrm{T}} \pmod{q}$$

which can be rewritten as:

$$\mathbf{f} \cdot \left(\mathbf{H}_h + \mathbf{v}^{\mathrm{T}}\left(\mathbf{w}^{-1}\right)^{\mathrm{T}}\right) \equiv l\left(\mathbf{w}^{-1}\right)^{\mathrm{T}} \pmod{q}$$

In conclusion, the original perfect hint

$$\langle(\mathbf{v}, \mathbf{w}), (\mathbf{f}, \mathbf{g})\rangle = l$$

can be transformed into a **secret-only mod-**q hint involving only $\mathbf{f}$, in the following form:

$$\mathbf{f} \cdot \left(\mathbf{H}_h + \mathbf{v}^{\mathrm{T}}\left(\mathbf{w}^{-1}\right)^{\mathrm{T}}\right) \equiv l\left(\mathbf{w}^{-1}\right)^{\mathrm{T}} \pmod{q}$$

This transformation allows the use of the *Modular Hints* method proposed in [29] to embed the hint into the NTRU lattice. As a result, the combined perfect hint involving both $\mathbf{f}$ and $\mathbf{g}$ can be effectively incorporated into the lattice structure, enabling the application of the Primal Lattice Reduction Attack to recover $\mathbf{f}$, and ultimately reconstruct the secret key sk.

4 Integrating Modular Hints in NTRU Instances

In this section, we focus on two common types of Modular Hints, namely the mod-q hint and the mod-2 hint that reflects low-bit leakage of secret key coefficients. Let

$$(\mathbf{H}_h, \mathbf{b}, q) \in \mathbb{Z}_q^{n\times m} \times \mathbb{Z}_q^m \times \mathbb{N}$$

denote an NTRU instance with secret polynomials f and g, and let $\bar{\mathbf{v}}_i = (\mathbf{v}_i, \ell_i, m_i) \in \mathbb{Z}^n \times \mathbb{Z} \times \mathbb{N}$ be a modular hint such that

$$\langle \mathbf{f}, \mathbf{v}_i\rangle \equiv \ell_i \mod m_i,$$

for $i = 1, \ldots, k$, where $\mathbf{s}$ denotes the secret.

For modular hints, we interpret them as noiseless RLWE samples reduced modulo m_i.

4.1 Integrating Mod-q Hints into NTRU Instances

In NTRU-based cryptographic schemes, the leakage of mod-q hint information is a common and significant security concern. For instance, during the key generation phase of the Falcon signature scheme, Number Theoretic Transform (NTT) operations are applied to the secret polynomial f. If such operations are subjected to side-channel analysis, they may result in the leakage of mod-q hint information on f.

The work of [29] focuses on the integration of *secret-only* mod-q hints, which, in the context of LWE, are of the form $\langle \mathbf{v}, \mathbf{s} \rangle \equiv \ell \mod q$. It further shows that more general *secret-error* hints—of the form $\langle (\mathbf{v}, \mathbf{w}), (\mathbf{s}, \mathbf{e}) \rangle \equiv \ell \mod q$—are equivalent to secret-only hints within each mod-q equivalence class. Therefore, without loss of generality, it suffices to consider only secret-only hints.

In the NTRU setting, the secret is given by a pair $(\mathbf{f}, \mathbf{g})$, and the core relation is:

$$\mathbf{f} \cdot \mathbf{H}_h - \mathbf{g} \equiv \mathbf{0} \mod q.$$

We consider two representative forms of joint leakage of $\mathbf{f}$ and $\mathbf{g}$ that may arise in practical attacks:

Type I (Combined Linear Leakage) This type takes the form:

$$\langle (\mathbf{f}, \mathbf{g}), (\mathbf{v}, \mathbf{w}) \rangle \equiv \ell \mod q, \quad \text{i.e.,} \quad \mathbf{f}\mathbf{v}^{\mathrm{T}} + \mathbf{g}\mathbf{w}^{\mathrm{T}} \equiv \ell \mod q.$$

By substituting the NTRU identity $\mathbf{g} \equiv \mathbf{f} \cdot \mathbf{H}_h \mod q$, the hint simplifies to a constraint purely on $\mathbf{f}$:

$$\mathbf{f}\left(\mathbf{v}^{\mathrm{T}} + \mathbf{H}_h \mathbf{w}^{\mathrm{T}}\right) \equiv \ell \mod q$$

Type II (Independent Component Leakage) In this case, the leaked mod-q hints on $\mathbf{f}$ and $\mathbf{g}$ are separate:

$$\langle \mathbf{f}, \mathbf{v} \rangle \equiv \ell_1 \mod q, \qquad \langle \mathbf{g}, \mathbf{w} \rangle \equiv \ell_2 \mod q.$$

Using $\mathbf{g} \equiv \mathbf{f} \cdot \mathbf{H}_h \mod q$, the second hint becomes:

$$\langle \mathbf{f}\mathbf{H}_h, \mathbf{w} \rangle \equiv \ell_2 \mod q,$$

.

Thus, both hints reduce to secret-only mod-q constraints on $\mathbf{f}$:

$$\langle \mathbf{f}, \mathbf{v} \rangle \equiv \ell_1 \mod q, \qquad \langle \mathbf{f}, \mathbf{H}_h \mathbf{w}^{\mathrm{T}} \rangle \equiv \ell_2 \mod q.$$

In conclusion, although the original leakage involves both $\mathbf{f}$ and $\mathbf{g}$, the equivalence class properties established in [29] allow us to convert such leakages into secret-only mod-q hints on $\mathbf{f}$. This not only simplifies the integration of leakage information but also enables efficient embedding into the NTRU lattice for conducting primal lattice reduction attacks, thereby enhancing the effectiveness of secret key recovery.

4.2 Integrating LSBs Hints into the NTRU Instances

Based on the NTRU cryptographic framework, we consider a common type of attack that may arise in practical scenarios. For instance, during the signing phase of the Hawk signature scheme, an adversary may be able to obtain the least significant bits (LSBs) of the secret key coefficients while computing the preimage, effectively resulting in *mod-2* hint leakage on the secret key.

Since the information revealed by LSBs is extremely limited, such leakage is generally insufficient to directly recover the secret key through practical attacks. The main objective of this section is to quantitatively evaluate the impact of such low-bit secret leakage on the security of the cryptographic scheme and to investigate how to simultaneously leverage the LSBs of both $\mathbf{f}$ and $\mathbf{g}$ in the attack context.

As for mod-2 hints, the work of [29] has discussed the case of *secret-only* modular hint leakage, where leakage of the form $\langle \mathbf{f}, \mathbf{v} \rangle \equiv \ell_1 \mod 2$ corresponds to a special instance of LSB-based modular hint leakage.

However, [29] does not cover scenarios where mod-2 hints on both $\mathbf{f}$ and $\mathbf{g}$ leak simultaneously. We identify two representative types of such joint leakage.

Type I (Combined Leakage) Hints are given in the form:

$$\langle (\mathbf{f}, \mathbf{g}), (\mathbf{v}, \mathbf{w}) \rangle \equiv \ell \mod 2.$$

Type II (Independent Component Leakage) Hints are given as:

$$\langle \mathbf{f}, \mathbf{v} \rangle \equiv \ell_1 \mod 2, \quad \langle \mathbf{g}, \mathbf{w} \rangle \equiv \ell_2 \mod 2.$$

For the NTRU setting, the core relation is:

$$\mathbf{f} \cdot \mathbf{H}_h - \mathbf{g} \equiv \mathbf{0} \mod q.$$

Assuming a Type-I joint mod-2 leakage and substituting $\mathbf{g} = \mathbf{f} \cdot \mathbf{H}_h \mod q$, we obtain the modular constraint:

$$\mathbf{f} \cdot \mathbf{H}_h \mathbf{w}^{\mathrm{T}} \equiv (\ell - \mathbf{f}\mathbf{v}^{\mathrm{T}} \mod 2) \mod q,$$

which can be directly embedded into the NTRU lattice for a primal attack.

In the case of Type-II leakage, i.e.,

$$\langle \mathbf{f}, \mathbf{v} \rangle \equiv \ell_1 \mod 2, \quad \langle \mathbf{g}, \mathbf{w} \rangle \equiv \ell_2 \mod 2,$$

substituting $\mathbf{g} = \mathbf{f} \cdot \mathbf{H}_h \mod q$ yields:

$$\left(\mathbf{f} \cdot \mathbf{H}_h \cdot \mathbf{w}^{\mathrm{T}} \bmod q\right) \bmod 2 \equiv \ell_2 \mod 2.$$

thus producing two independent modular constraints on $\mathbf{f}$:

$$\langle \mathbf{f}, \mathbf{v} \rangle \equiv \ell_1 \mod 2, \quad \left(\mathbf{f} \cdot \mathbf{H}_h \cdot \mathbf{w}^{\mathrm{T}} \bmod q\right) \bmod 2 \equiv \ell_2 \mod 2.$$

Both of these hints can be simultaneously embedded into the NTRU lattice to perform the primal lattice reduction attack.

LSBs Analysis for Falcon-512 We evaluate the leakage of least significant bit (LSB) information of secret key coefficients in Falcon-512, modeling the LSB leakage of f as a modular constraint of the form *mod-2 hint*. The evaluation in this section is performed using the `leaky-LWE-Estimator` tool proposed in [8].

Prior to the integration of any auxiliary hint information, the original primal lattice reduction instance has a dimension of $n = 1025$, and successful key recovery in this setting requires a BKZ lattice reduction with blocksize $\beta = 350.46$. It is important to note that short vector hints are regarded as intrinsic components of the lattice structure and are typically integrated after external hints such as modular hints.

In the scenario where only mod-2 hints are embedded, the lattice dimension involved in the primal attack remains unchanged at 1025. However, when short vector hints are additionally incorporated on top of the mod-2 leakage, a reduction in the effective lattice dimension is observed, along with a corresponding decrease in the required BKZ blocksize.

We provide a detailed tabulation of the remaining lattice dimension (d) and the corresponding optimal BKZ blocksize under the combined setting of mod-2 hints and short vector hints. This evaluation supports a fine-grained analysis of the impact that different types of hint combinations have on the complexity of lattice attacks, and contributes to a deeper understanding of the synergistic effects of leakage and their implications for the overall security degradation of lattice-based cryptographic schemes.

Table 4: Lattice dimension, blocksize variation, and time cost under different numbers of Mod-2 hints

Number of Mod-2 hints	460	470	480	490	500	510	512
Dimension d	988	987	985	984	983	981	981
Δd	37	38	40	41	42	44	44
Remaining β	320.77	320.14	319.50	318.87	318.23	317.60	317.47
$\Delta\beta$	29.69	30.32	30.96	31.59	32.23	32.86	32.99
Time cost (h)	9.28	9.51	9.74	9.72	10.28	10.24	10.34

LSBs Analysis for Hawk-512 In the Hawk signature scheme, certain intermediate computational results, such as precomputations, are essentially equivalent to providing modular hints of f, F, g, G modulo 2.

Hawk signatures also suffer from Hamming weight information leakage. Obtaining the Hamming weight of the string kgseed represented as $(b_1 b_2 \dots b'_1 b'_2 \dots)$ may expose sensitive information. In the Hawk signature scheme, the secret keys f and g are generated by sampling from a centered binomial distribution using bit strings, which leads to:

$$f = \sum b_i - b_i', \quad g = \sum b_i - b_i'$$

where b_i, b_i' are binary bits. Since the Hamming weight is relatively easy to leak, it directly reveals the number of ones. Therefore, some bit-level information (e.g., $f \mod 2, g \mod 2$) may be more easily exposed compared to the complete Hamming weight. This method of obtaining Hamming weight can be leveraged to construct template-based side-channel attacks.

We now analyze the impact of obtaining the least significant bit information of f, F, g, G through side-channel attacks on lattice reduction attacks, which means inserting the information of Modular-hint into the lattice reduction attack, equivalent to sparsifying the lattice Λ^T. The equation can be rewritten as:

$$\begin{pmatrix} t_0 \\ t_1 \end{pmatrix} = \begin{pmatrix} f & F \\ g & G \end{pmatrix} \begin{pmatrix} h_0 \\ h_1 \end{pmatrix} \mod 2 \tag{10}$$

where

$$\mathbf{B} = \begin{pmatrix} f & F \\ g & G \end{pmatrix}$$

is considered as secret key information, and we apply $4n$ **modular-hints**. The estimation script in the previous section only takes the lattice dimension and volume as inputs.

To assess the security loss caused by $4n$ modular-hints, we can use [8] Lemma 13 and deduce that, under the primitivity hypothesis of h, each hint increases the volume by a factor of 2.Thus, the overall volume is expanded by 2^{4n}.

In listing 2, a script is provided for looking up these values.

```
for n in {512, 1024, 2048}:
    beta, preV_sd= predict_beta_and_prev_sd(n, n*log(n)/2 + n/2,
                   lift_union_bound=True, number_targets=n, tours=1)
    print(beta)
```

Listing 1: Estimation of the residual complexity with the knowledge of LSB of $\mathbf{f}$.

```
for n in {512, 1024, 2048}:
    beta, preV_sd= predict_beta_and_prev_sd(n, n*log(n)/2 + 2n,
                   lift_union_bound=True, number_targets=n, tours=1)
    print(beta)
```

Listing 2: Estimation of the residual complexity with the knowledge of LSB of $(\mathbf{f}, \mathbf{F}, \mathbf{g}, \mathbf{G})$.

In the function, the second parameter `logvol` refers to the logarithm of the lattice volume. Originally, the volume in the code is given by:

$$V = \frac{n \log n}{2}$$

Since there are n hints, we can directly add n to the logarithm of the volume, resulting in the updated volume:

$$V = \frac{n \log n}{2} + \frac{n}{2}$$

Thus, when $4n$ hints are leaked, the volume becomes:

$$V = \frac{n \log n}{2} + 2n$$

where $(\mathbf{f}, \mathbf{F})$ are target vectors and also part of the private key. The secret key $\mathbf{sk}$ can be recovered through $(\mathbf{f}, \mathbf{F})$.

$$\begin{pmatrix} \mathbf{t}_0 \\ \mathbf{t}_1 \end{pmatrix} = \begin{pmatrix} \mathbf{f} & \mathbf{F} \\ \mathbf{g} & \mathbf{G} \end{pmatrix} \begin{pmatrix} \mathbf{h}_0 \\ \mathbf{h}_1 \end{pmatrix} \mod 2 = \begin{pmatrix} \mathbf{f} \cdot \mathbf{h}_0 + \mathbf{F} \cdot \mathbf{h}_1 \\ \mathbf{g} \cdot \mathbf{h}_0 + \mathbf{G} \cdot \mathbf{h}_1 \end{pmatrix} \mod 2$$

$$\begin{pmatrix} \mathbf{x}_0 \\ \mathbf{x}_1 \end{pmatrix} = \begin{pmatrix} \mathbf{f} & \mathbf{F} \\ \mathbf{g} & \mathbf{G} \end{pmatrix} \begin{pmatrix} \mathbf{w}_0 \\ \mathbf{w}_1 \end{pmatrix}$$

It is known that the value of $\mathbf{x}_0 \mod 2$ is actually the same as the values of $\mathbf{t}_0$ and $\mathbf{t}_1$ above.

In [17], the primary discussion focuses on the impact of partial information leakage of the discrete Gaussian sampling stage $\mathbf{x}$ in the Hawk signature scheme on solving the target short vectors $(\mathbf{f}, \mathbf{F})$. However, the leakage of preimage $\mathbf{t}$ in discrete Gaussian sampling also affects the complexity of solving and recovering the target short vectors $(\mathbf{f}, \mathbf{F})$ and $(\mathbf{g}, \mathbf{G})$, as well as recovering the private key.

The leakage of preimage $\mathbf{t}$ primarily occurs through integer multiplications in side-channel attacks. Specifically, in step 8 of the signature algorithm, CPA (Correlation Power Analysis) attacks in side-channel analysis can extract the least significant bit (LSB) of each coefficient in f, g, F, G.

The leakage of preimage $\mathbf{t}$ results in partial private key leakage, specifically revealing the LSB of each coefficient of f, g, F, G.

Table 5: Estimated residual beta obtained with the knowledge of $(\mathbf{f}, \mathbf{F}, \mathbf{g}, \mathbf{G})_{\text{LSB}}$.

β	No information	$\mathbf{f}_{\text{LSB}}$	$\mathbf{F}_{\text{LSB}}$	$\mathbf{g}_{\text{LSB}}$	$\mathbf{G}_{\text{LSB}}$	$(\mathbf{f}, \mathbf{F}, \mathbf{g}, \mathbf{G})_{\text{LSB}}$	Hint type
HAWK-256	211	172	172	172	172	$211 - 39 \times 4 = 55$	Modular hint
HAWK-512	452	387	387	387	387	$452 - 65 \times 4 = 192$	Modular hint
HAWK-1024	940	829	829	829	829	$940 - 131 \times 4 = 416$	Modular hint

For Hawk-256, the leakage of preimage $\mathbf{t}$ reveals 1024 modular hints, reducing the LWE problem dimension by $39 \times 4 = 156$. For Hawk-512, the leakage of preimage $\mathbf{t}$ reveals 2048 modular hints, reducing the LWE problem dimension by $65 \times 4 = 260$. For Hawk-1024, the leakage of preimage $\mathbf{t}$ reveals 4096 modular hints, reducing the LWE problem dimension by $131 \times 4 = 524$.

The required block size without information leakage and the remaining block size after the leakage of the LSB of each coefficient of f, g, F, G are shown in the table below.

5 Experimental Analysis

5.1 Experimental environment

All experiments were conducted on a server running Ubuntu 20.04.6 LTS with a 5.15.0-130-generic Linux kernel (x86_64 architecture). The server is equipped with two Intel(R) Xeon(R) Gold 5220R CPUs @ 2.20GHz, providing a total of 48 physical cores and 96 logical threads. The system is configured with 503 GiB of main memory and a 122 GiB swap space. For GPU acceleration, the server hosts three NVIDIA RTX A4000 graphics cards, each equipped with 16 GiB of dedicated memory. The GPUs operate under CUDA version 12.4 and NVIDIA driver version 550.54.15. The computational experiments were primarily executed using SageMath version 10.4, which was compiled from source to ensure compatibility with the server environment.

5.2 Experimental Setup

In our experiments, we consider the case of **perfect hints**, where some coefficients of the secret polynomials f and g are assumed to be leaked. The hint can be represented as an inner product of the form $\langle \mathbf{v}_i, \mathbf{f} \rangle = \ell_i$, where $\mathbf{v}_i$ can be interpreted as a standard basis vector. For example, the i-th standard basis vector is given by $\mathbf{e}_i = (0, \ldots, 1, \ldots, 0)$.

For modular hints, we also focus on the leakage of partial coefficients. In particular, we investigate two commonly used types in practical implementations: (1) the Mod-q hint, which reveals relatively more information, expressed as $\langle \mathbf{v}_i, \mathbf{f} \rangle = \ell_i \bmod q$ and the Mod-2 hint, which corresponds to the most limited form of leakage, revealing only low-order bits, expressed as $\langle \mathbf{v}_i, \mathbf{f} \rangle = \ell_i \bmod 2$.

Under each type of hint, we conduct experiments on 5 different NTRU instances for each signature scheme. We assess the security complexity of the primal lattice reduction attack by the minimal BKZ blocksize β required to achieve key recovery, and the time complexity by the corresponding clock time needed to complete the attack.

Our experimental analysis is based on existing efficient open-source Python and Sage implementations. The source code and comprehensive documentation are available at: `https://github.com/juliannowakowski/lwe_with_hints`, `https://github.com/lducas/leaky-LWE-Estimator`. The Falcon and Hawk related experiments in this paper are based on the open source reference implementations provided by NIST: `https://github.com/tprest/falcon.py` and `https://github.com/hawk-sign/aux/tree/main/code`.

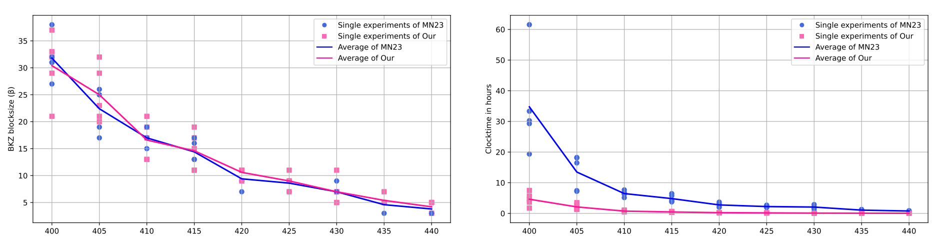
5.3 Experimental Results for secrect-only f perfect hint

perfect hint for Falcon In the context of partial coefficient leakage of the private key vector f (i.e., perfect hints), we selected five NTRU instances derived from Falcon signatures to perform experimental analysis. The discrete points in the figure represent the results of individual attacks on these instances, while the solid lines represent the average performance.

Figure 1a presents the variation in the BKZ blocksize β required for a successful attack under different levels of known coefficients. The results indicate that our method yields security degradation comparable to that of the MN23 approach, as both exhibit nearly identical trends and numerical values, thereby validating the effectiveness of our modeling.

Figure 1b further illustrates the comparison of average attack clocktimes between the two methods during the key recovery process. At perfect hint counts of 400 and 405, under average conditions, our method requires only 6 hours and 2 hours, respectively, whereas the MN23 method incurs clocktimes of approximately 35 hours and 15 hours. This corresponds to clocktime reductions of approximately 5.83× and 7.5×, respectively.

Overall, our approach preserves comparable attack success rates while significantly lowering computational cost, thus demonstrating enhanced efficiency and practical applicability. These findings provide compelling evidence that our improved modeling framework delivers superior performance for security evaluation under coefficient leakage scenarios. The two figures below illustrate, for Falcon-512, the required BKZ blocksize and clocktime to recover the full secret key f, given the knowledge of its coefficients from index 400 to 440.



(a) Effect of PSCL of f on BKZ blocksize in Falcon-512.

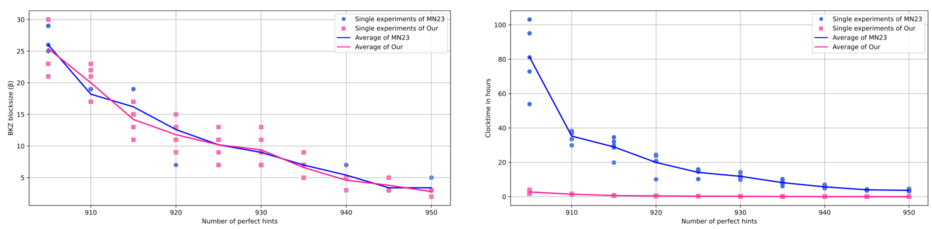
(b) Effect of PSCL of f on attack clocktime in Falcon-512.

Fig. 1: Effect of perfect secret coefficient leakage (PSCL) of f on the BKZ blocksize and clocktime in Falcon-512.

Figure 2b illustrates the clock time required to recover the Falcon-1024 secret key under varying numbers of leaked perfect hints. When 910 coefficients of f are leaked, our proposed method achieves full key recovery with an average clock time of less than 2 hours, while the MN23 approach requires over 30 hours on average, with some individual runs exceeding 35 hours.

As the number of leaked coefficients increases to 950, our method exhibits a slight improvement, reducing the average recovery time to below 1 hour. In contrast, MN23's performance also improves but remains substantially slower, requiring approximately 5 hours on average. The gap in efficiency is therefore especially pronounced when the leakage is close to minimal: at 910 hints, our method is over 15× faster on average, whereas at 950 hints, it remains about 5× faster.

These results demonstrate the scalability and efficiency of our attack framework under high-dimensional NTRU parameter sets, especially in scenarios with sparse or borderline leakage.

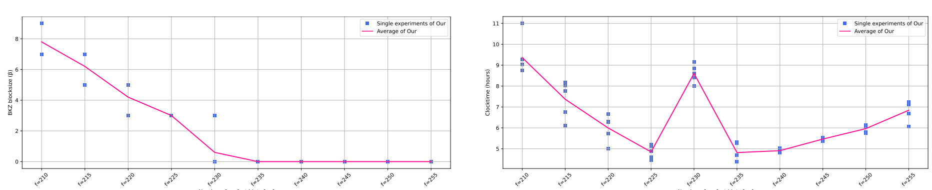


(a) Effect of PSCL of f on BKZ blocksize in Falcon-1024.

(b) Effect of PSCL of f on attack clocktime in Falcon-1024.

Fig. 2: Effect of perfect secret coefficient leakage (PSCL) of f on the BKZ blocksize and clocktime in Falcon-1024.

We conduct practical attacks on Mitaka-512 using the method of integrating perfect hints proposed in [29], under the scenario where the secret $\mathbf{f}$ is partially leaked in the form of perfect hints. The figures illustrate the relationship between the number of perfect hints and the BKZ blocksize as well as the clock time (in hours) required for the attack on Mitaka-512.



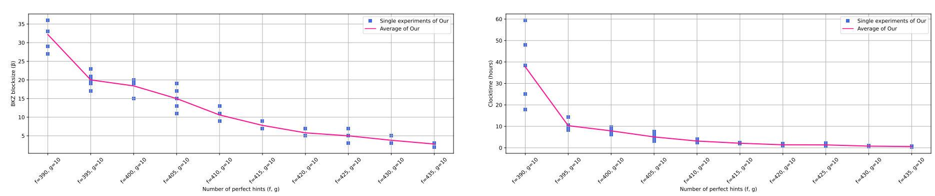
(a) Effect of perfect hints of f on BKZ blocksize in Mitaka-512.

(b) Effect of perfect hints of f on attack clocktime in Mitaka-512.

Fig. 3: Impact of perfect secret coefficient leakage of f on BKZ blocksize and clocktime in Mitaka-512.

5.4 Experimental Results for secrect f and g perfect hint

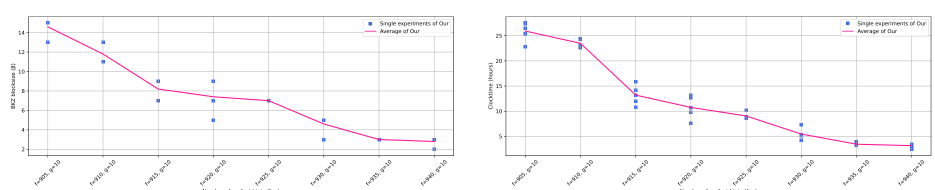
perfect hint for Falcon By integrating partial known coefficients (i.e., perfect hints) of the private key vectors $\mathbf{f}$ and $\mathbf{g}$ into a unified lattice structure, we can evaluate the residual security strength of the Falcon scheme under side-channel information leakage. The remaining security level is quantified by the required BKZ blocksize β. Concurrently, we embed these hint constraints into a modified sublattice and conduct a primal lattice reduction attack to recover the secret key, where the attack cost is measured in clocktime (hours). Experimental results demonstrate that as the number of leaked coefficients increases, the required BKZ blocksize decreases significantly, and the attack time is reduced exponentially. This clearly reveals the security degradation caused by the leakage of f and g's coefficients. Moreover, these results quantitatively illustrate the time complexity for recovering the Falcon secret key given partial information, providing empirical support for assessing security strength under various leakage scenarios.



(a) Effect of PSCL of f and g on BKZ blocksize in Falcon-512.

(b) Effect of PSCL of f and g on clocktime in Falcon-512.

Fig. 4: Effect of perfect secret coefficient leakage (PSCL) of f and g on the BKZ blocksize and attack clocktime in Falcon-512.



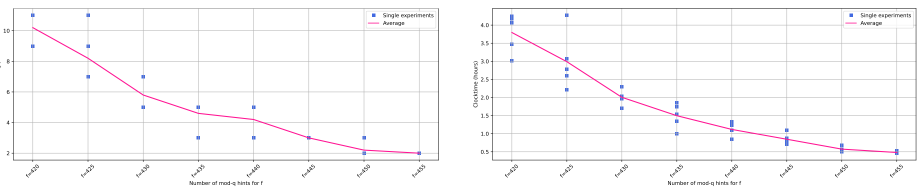
(a) Effect of PSCL of f and g on BKZ blocksize in Falcon-1024.

(b) Effect of PSCL of f and g on clocktime in Falcon-1024.

Fig. 5: Impact of perfect secret coefficient leakage (PSCL) of f and g on BKZ blocksize and clocktime in Falcon-1024.

5.5 Experimental Results for Modular hint of f

We conduct practical attacks on Mitaka-512 using the method of integrating mod-q hints proposed in [29], under the scenario where the secret f is partially leaked in the form of mod-q hints. The figures illustrate the relationship between the number of mod-q hints and the BKZ blocksize as well as the clock time (in hours) in Mitaka-512.



(a) Effect of mod-q hints of f on BKZ blocksize in Mitaka-512.

(b) Effect of mod-q hints of f on attack clocktime in Mitaka-512.

Fig. 6: Impact of mod-q hint leakage of the secret f on BKZ blocksize and clocktime in Mitaka-512.

6 Conclusion and Future Work

Our work investigates the impact of partial information leakage on the security of NTRU-based schemes. We begin by analyzing how different types of perfect hints affect the lattice volume. Subsequently, we propose an efficient attack methodology that enables full secret key recovery under partial key leakage in structured NTRU lattices. We further examine how to evaluate security and conduct practical attacks in scenarios where partial coefficients and least significant bits of both f and g are leaked. Finally, we perform a comprehensive security evaluation and practical key recovery attacks on Falcon, Mitaka, and Hawk under various forms of secret key leakage.

In future work, we plan to explore the following three research directions. First, we aim to investigate the efficient integration of approximate hints, focusing on their impact on the overall security degradation and the feasibility of full secret key recovery under such leakage. Second, motivated by the recent work of Hermelink et al. [18], who proposed a unified framework that models various types of secret key leakage as distributional hints and solves the resulting recovery problem using greedy approach or belief propagation, we intend to study the applicability and effectiveness of this approach in the context of NTRU-based constructions.

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